

Predicting Factor Decay: ML Models for Cross-Factor Predictability Erosion

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Equity factors and predictive relationships

Six daily factor series define 30 directed relationships over 1990–2024.

Source information

HML captures value relative to growth. Past HML returns may help forecast another factor.

Forecast target

SMB captures small relative to large firms. The target is its later return, conditional on its own history.

Meaning of a relationship

Granger tests ask whether source lags add predictive content. They do not identify a structural causal effect.

- The six-factor panel combines Fama–French five factors with momentum.
- The paper reports 8,817 trading days.

Regimes, predictive tests, and decay models

The regime labels describe fitted distributions, not fixed economic event categories.

Regime discovery

Student-t hidden Markov models group return behavior into three statistical states. Fifty random starts assess sensitivity to local optima.

Relationship measurement

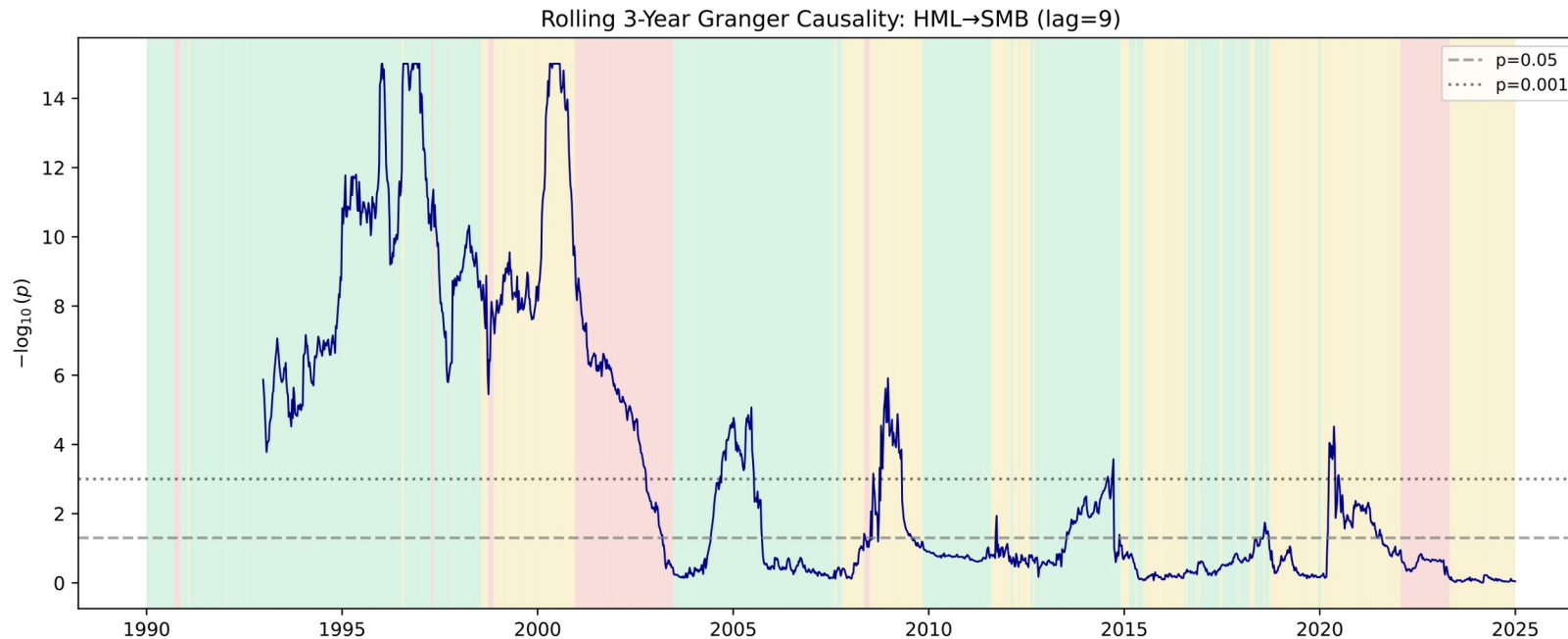
Within each state, Granger tests measure source-lag contribution. Rolling statistics track how that evidence changes.

Decay prediction

Classifiers predict a loss of significance within five years. Survival models rank the time until a decay event.

Predictive evidence changes over time

The paper's rolling analysis shows episodic changes in value-to-size predictability.



- The plotted quantity is $-\log_{10}(p)$, not a trading return.
- Nine lags describe the included history, not nine-day advance warning.

Original manuscript figure: rolling three-year unconditional Granger evidence, using lags 1–9 and regime-colored background.

Learning which relationships may weaken

Each observation is a directed factor pair. F summarizes evidence that source lags improve prediction.

- Train on 20 pairs and evaluate on 10 held-out pairs.
- Use initial F strength, rolling F behavior, correlation, and relative volatility.
- A binary event means F loses significance within five years.
- Survival analysis models event timing, with seven events among thirty pairs.

Strong CV ranking, limited holdout evidence

Only seven decay events support the reported model comparison.

Evaluation	Metric	Reported result
Random Survival Forest, 5-fold CV	C-index	0.933 ± 0.097
Logistic regression, 5-fold CV	AUC	0.933 ± 0.133
Separate ten-pair holdout	Accuracy	60% (6/10)

- C-index evaluates the ordering of event times.
- The CV standard deviation is not a confidence interval.
- Pair-level validation does not establish future-time deployment performance.

Regime detection has a speed-accuracy tradeoff

Detection delay measures recognition of a regime change after it occurs.

Method	Delay	Accuracy	F1
Online HMM	0.2 days	50.5%	0.22
Monthly rolling HMM	6.6 days	47.6%	0.39
Volatility threshold	18.3 days	53.0%	0.48

- The fastest method has the lowest reported F1.
- These numbers concern regime classification, not factor-decay forecasting.

The US relationship fails corrected OOS tests

The HMM is fitted on 1990–2012 and frozen for the 2013–2024 evaluation.

Regime	HAC p-value	Incremental R ²
Normal	0.152	0.32%
Elevated	0.041	0.73%
Crisis	0.290	0.09%

- Elevated $p = 0.041$ exceeds the 0.0167 correction for three regimes.
- The result is also sensitive to bandwidth and regime prevalence.
- Failure to reject does not prove that predictive information is absent.

Cross-market effects vary by region and state

The two crisis-state results survive the reported 12-comparison threshold of 0.0042.

Selected crisis-state effects

Developed ex-US and Asia-Pacific ex-Japan show HAC $p < 0.001$ in the reported later-period crisis-state tests.

Limits to generalization

Europe and Japan show in-sample evidence but later-period nulls. Regional significance is not evidence of a universal decay predictor.

Signal durability remains an exploratory target

The contribution is a framework for studying when measured predictability weakens.

- Seven events and shared factors limit precision and independence.
- Some pre-2012 decay fits use full-sample regime assignments.
- Post-hoc pair selection and model choices complicate inference.
- A fully prospective study would refit all stages using only available history.

Source: github.com/ChorokLeeDev/factor-decay-icaif2026